```
...
            }}
```

❶ Data structure for each student participant

❷ Mapping from account address to student structure

❸ Event definitions

❹ Modifier definitions

❺ Functions for adding a student user

❻ Functions for adding courses

❼ Function for determining certificate eligibility

CODING THE FUNCTIONS

Download the DCC.sol from the codebase, and review it. Observe the simplicity of the data structures and the functions. I'll discuss just two snippets:

- Adding a prerequisite course (115 or 116)
- Checking the eligibility of the prerequisite course requirement of the DCC

Here is the function for adding a prerequisite course:

```
function addPreRequisiteCourse(uint courseNumber, uint grade) public validStudent
{
        if(courseNumber == 115) {
            registeredStudents[msg.sender].prereq115 = grade;
        }
        else if(courseNumber == 116) {
            registeredStudents[msg.sender].prereq116 = grade;
        }
        else {
            revert("Invalid course information provided");
        }
```

In the header of the function, the modifier `validStudent` enforces the condition that the caller of this function (sender of the Tx: msg.sender) must have a registered identity. The parameters passed to the function are the course number and the grade for that course. The body of the functions checks the